

TITLE PAGE

- Problem Statement ID – **SIH25067**
- Problem Statement Title-
Proposal for Design and development of application or Heavy Metal Pollution indices
- Theme- **Miscellaneous**
- PS Category- **Software**
- Team ID- **74274**
- Team Name-  **PRAVAH**

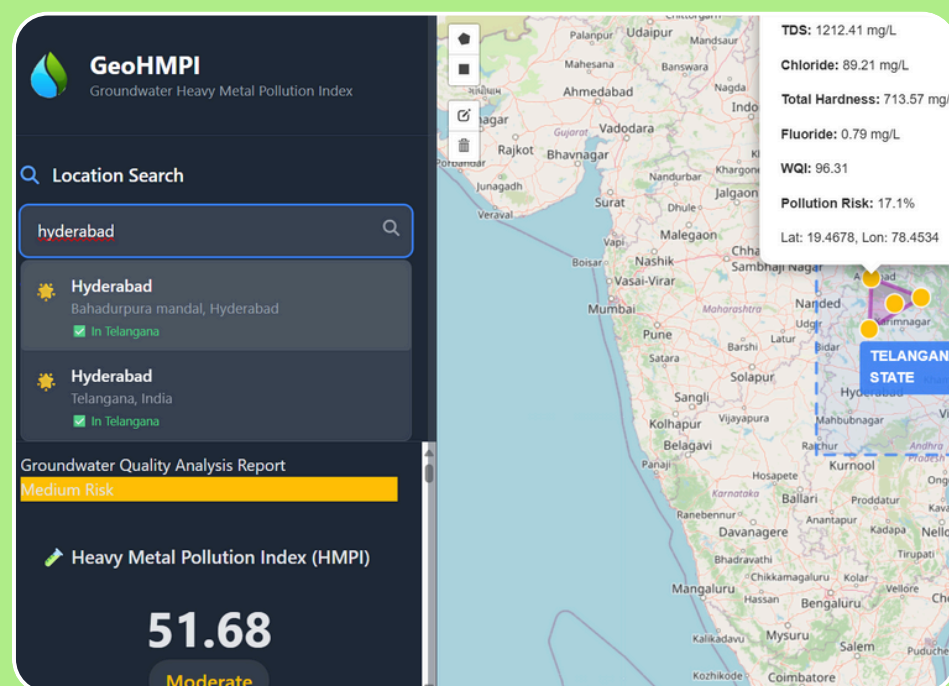


The Problem

Groundwater, the main drinking water source for over **35 million** people, faces a critical threat from heavy metal contamination.

Metals like lead in drinking water can cause **cancer**, yet current monitoring **lacks real-time prediction** and **automated Heavy Metal Pollution Indexing (HMPI)**, limiting proactive detection of contamination hotspots.

Proposed Solution: **GeoHMPI** - A website for Groundwater Heavy Metal Pollution Index Analysis



1. Location-based 8 Heavy-Metal Concentration prediction using ML

Multi-Metal Prediction
(Using neighbouring location data):

- Iron
- Manganese
- Copper
- Zinc
- Lead
- Cadmium
- Chromium
- Nickel

Plus calculating **8 other water quality indicators**:

1. **TDS**: Total Dissolved Solids
2. **SAR**: Sodium Adsorption Ratio
3. **WQI**: Water Quality Index, etc..

2. Automatic Heavy Metal Pollution Index (HMPI) Calculation

WHO/BIS standard-based HMPI calculation:

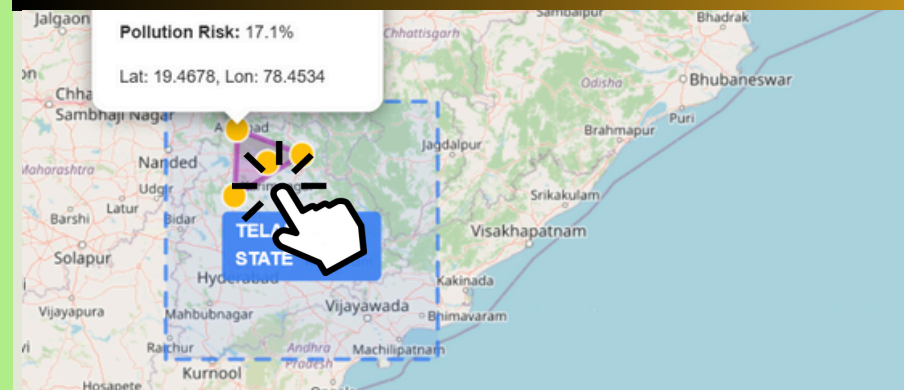
$$HMPI = \frac{\sum(W_i \times Q_i)}{\sum(W_i)}$$

- **Q_i** = Quality Rating = $(C_i / S_i) \times 100$
- **W_i** = Toxicity Weight for parameter i
- **C_i** = Measured Concentration of parameter i
- **S_i** = WHO/BIS Standard for parameter i

3. Multi-Location HMPI Analysis:

Bulk data analysis with **CSV** File upload

4. Interactive Geospatial Web Platform



Map Click-to-Predict Interface:

Real-time pollution assessment by clicking anywhere on Telangana map
(For now, model trained only on Telangana data)

Machine Learning Pipeline:

- Used Ensemble models (Random Forest, Gradient Boosting)
- Trained on 3 years **Telangana** data (2018-2020) (for now)

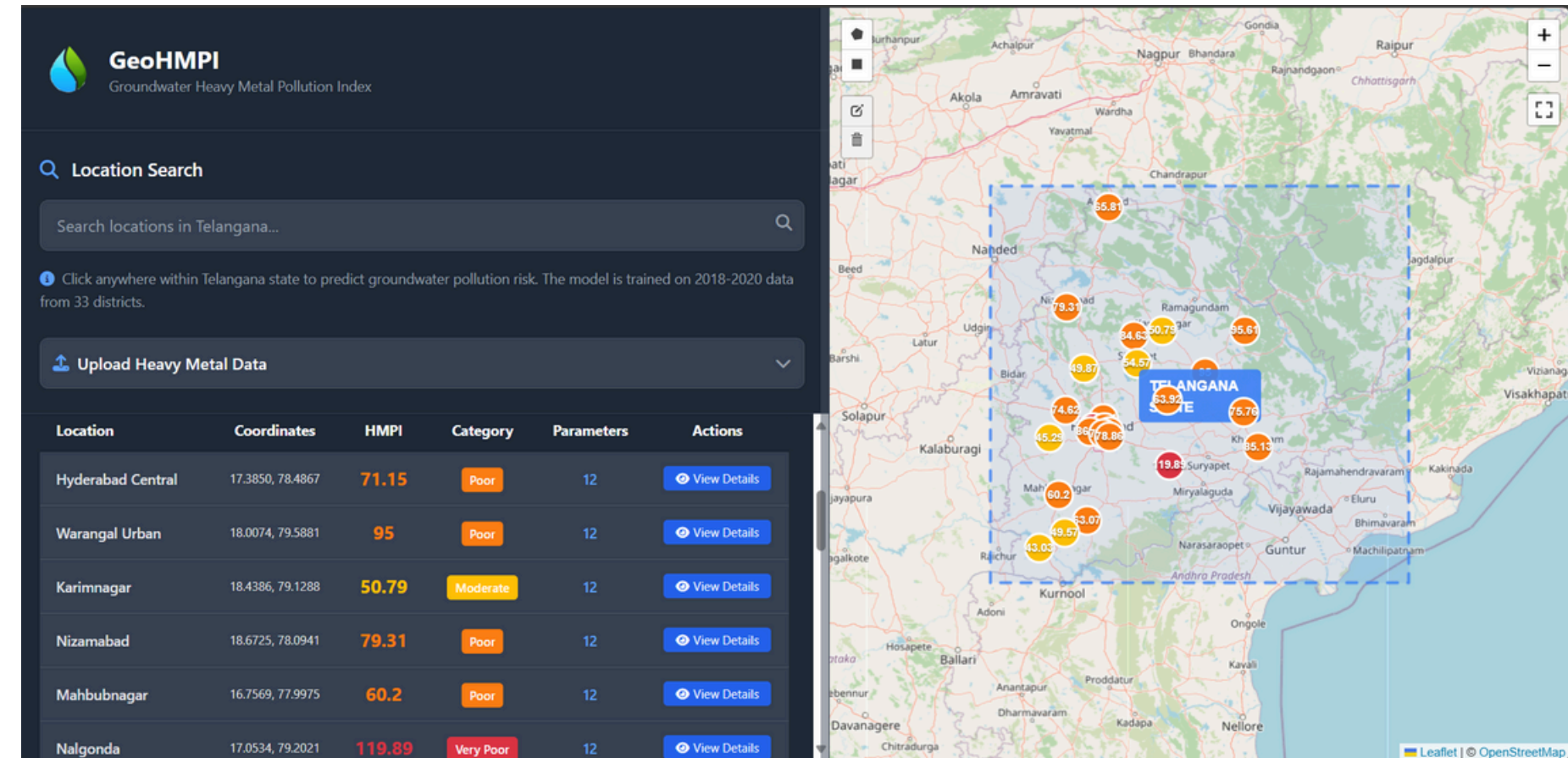
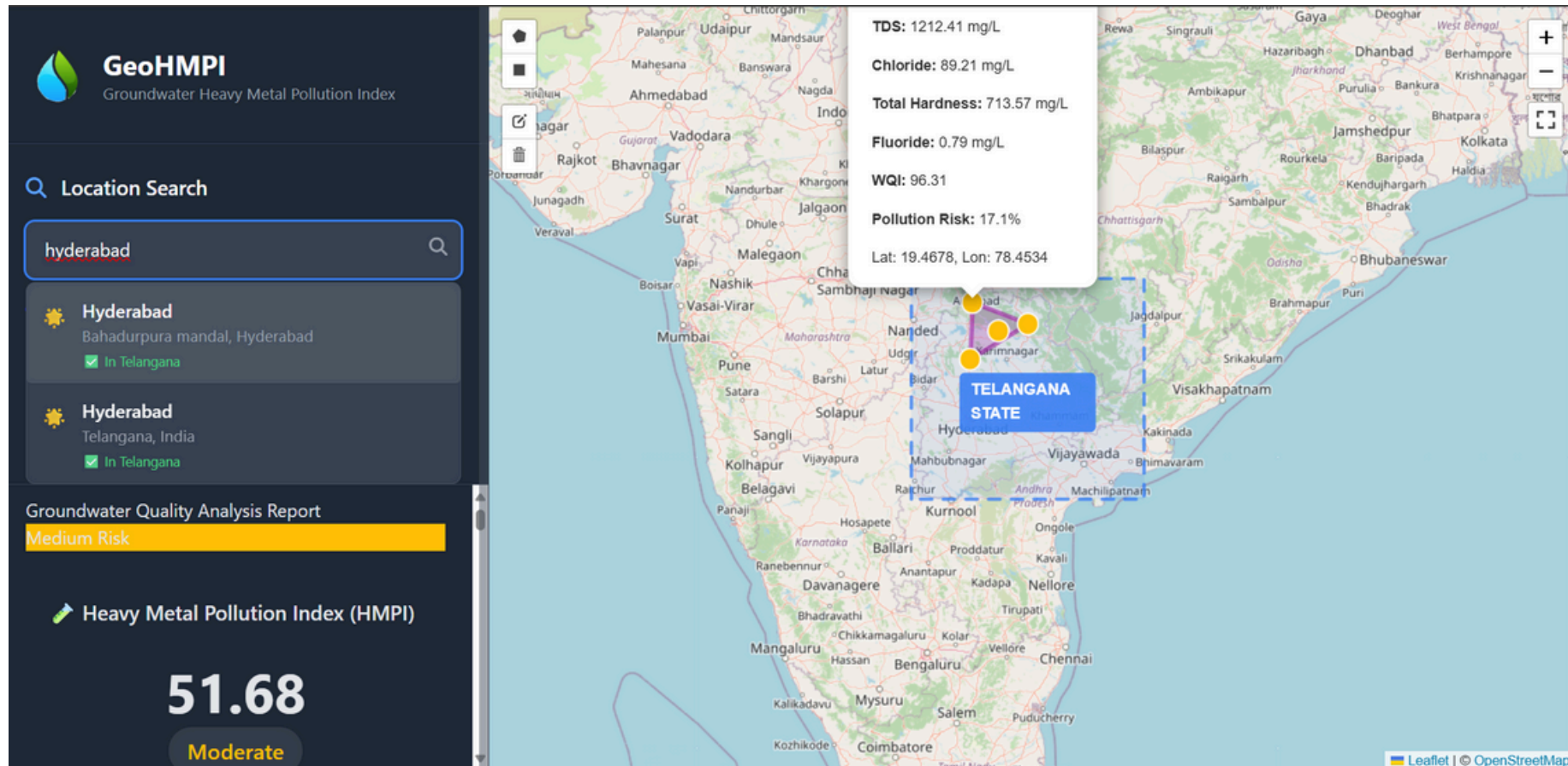
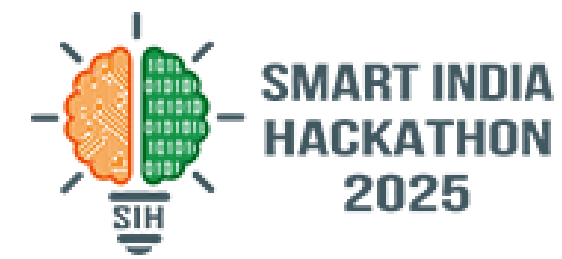
HMPI Classification Categories:

$15 \leq \text{HMPI} < 30$: Good (Generally safe for consumption)

$60 \leq \text{HMPI} < 100$: Poor (Requires treatment)



PROPOSED PROTOTYPE



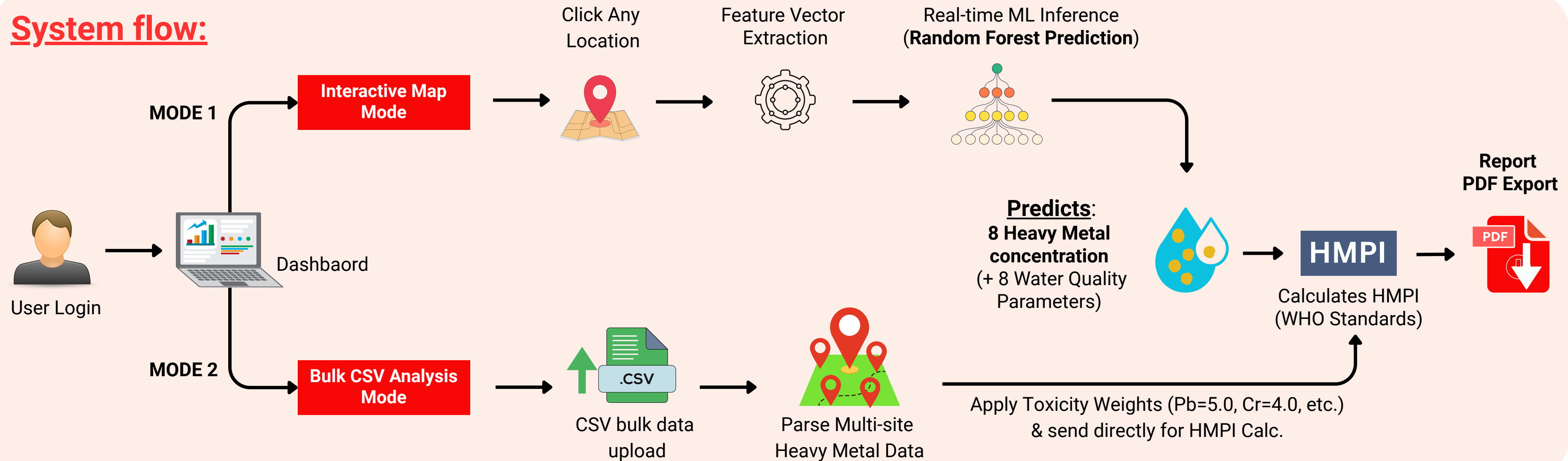
MODE 1: HMPI Analysis of **one location** selected on **interactive map** by user

MODE 2: CSV File HMPI Analysis with Heavy Metal Data of **multiple location** in Telangana

For prototyping, we got the **Ground Water Quality Dataset** for only **Telangana** State from **Kaggle**. In future, we will expand to other areas as well.

Training Data:
2018-2020 Telangana groundwater quality dataset with 16 features and 8 target parameters across 33 districts.

System flow:



Tech Stack:



Python Web Framework



Interactive Mapping



Machine Learning Library

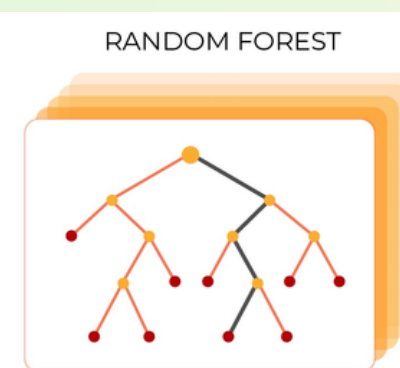


Frontend Interface



Data Processing

Machine Learning Approach:



We chose **Random Forest** because it gives **stable and robust** predictions, especially with **noisy and non-linear environmental data**.

Toxicity Weights

High Toxicity Metals:

- Lead (Pb): 5.0
- Cadmium (Cd): 5.0

Moderate Toxicity:

- Zinc (Zn): 2.0, etc.



Feasibility:

1. **Technical:** Uses proven Flask, Scikit-learn, Leaflet.js technologies.
2. **Data:** Built on 3 years validated Telangana groundwater data.
3. **Infrastructure:** Lightweight web app with minimal server requirements.



Use Cases:

1. Environmental consultants for rapid preliminary assessments
2. Rural communities accessing professional water quality evaluation
3. Researchers processing large geographic datasets
4. Government agencies conducting regional surveys



Viability:

1. **Cost-effective:** Reduces ₹10,000 lab testing to free predictions
2. **Scalable:** Handles individual queries and bulk processing
3. **Market demand:** Addresses affordable water quality assessment gap



Challenges:

1. **Accuracy limitations:** 85-92% vs 95-99% laboratory precision.
2. **Regional scope:** Currently limited to Telangana dataset.
3. **User adoption:** Training professionals on digital assessment tools.
4. **Regulatory acceptance:** Validation for official compliance reporting.



Solutions:

1. Position as preliminary screening tool, not replacement for critical testing
2. **Expand dataset collection to other states and regions**
3. Partner with certified labs for official validation processes



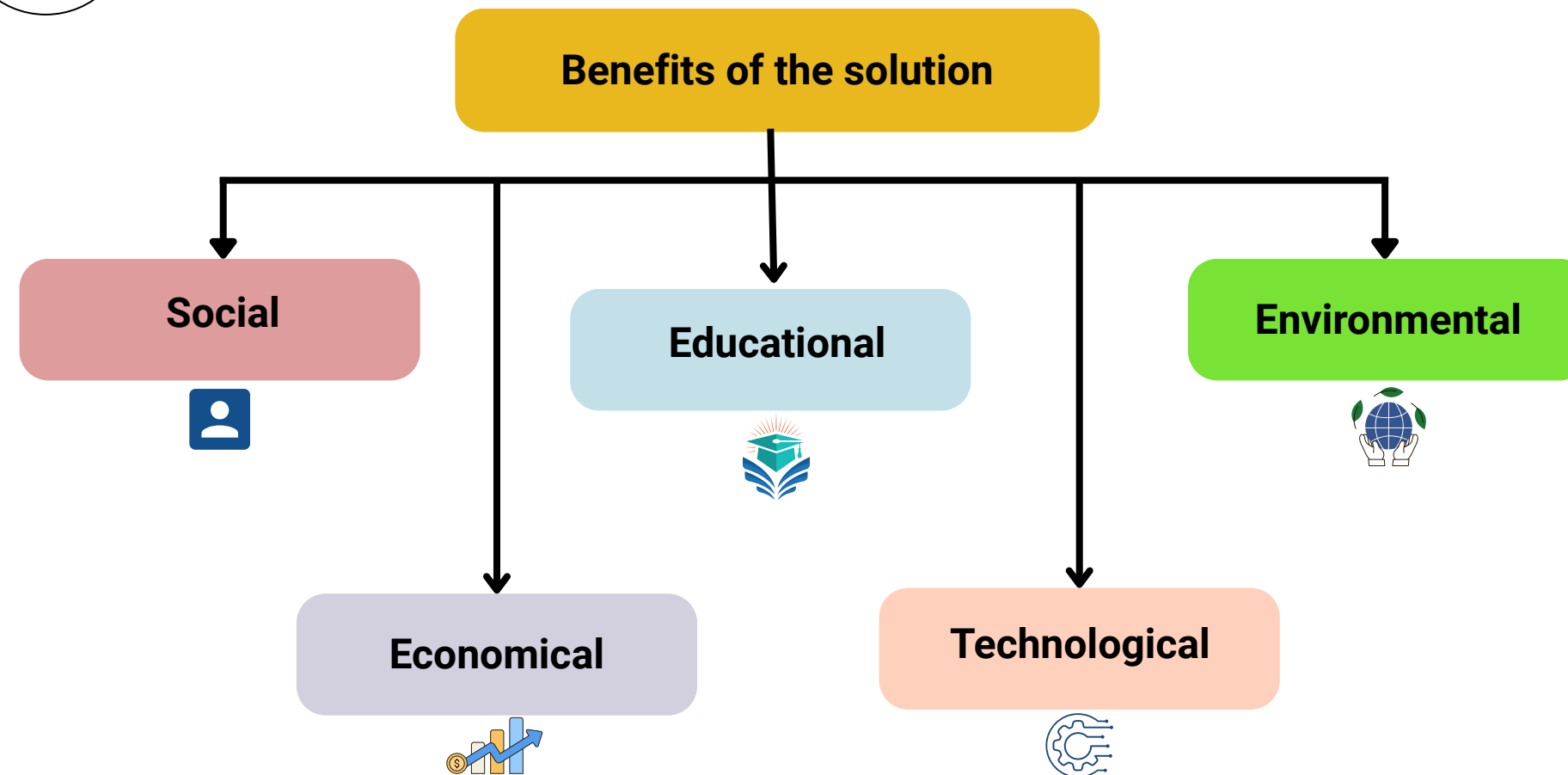
Implementations:

1. **Phase 1:** Deploy for Telangana with current ML models and dataset
2. **Phase 2:** Expand to neighboring states through data collection partnerships
3. **Phase 3:** Develop mobile application for field-based assessments
4. **Phase 4:** Integrate with **IoT sensors** for real-time monitoring systems



Supporting Facts:

- Processes 1000+ locations in minutes vs traditional 3-7 days per sample
- Built with standard web technologies ensuring maintainability
- Shows 85-92% accuracy against real groundwater data



❖ Impact on Target Audience:

- **Environmental consultants:** Complete preliminary assessments in seconds instead of weeks, enabling faster project delivery
- **Government agencies:** Monitor entire districts instantly through bulk CSV processing, improving regional surveillance
- **Research institutions:** Process large hydrogeological datasets efficiently for comprehensive contamination studies

❖ Benefits of the solution:

Social:

- Instant water quality assessment for communities lacking laboratory access
- Real-time contamination detection protecting public health

Economic:

- Eliminates ₹10,000 laboratory testing costs through free digital predictions
- Reduces environmental assessment time from 7 days to 2 seconds

Educational:

- Demonstrates practical ML application in environmental problem-solving
- Interactive platform for studying groundwater contamination patterns

Technological:

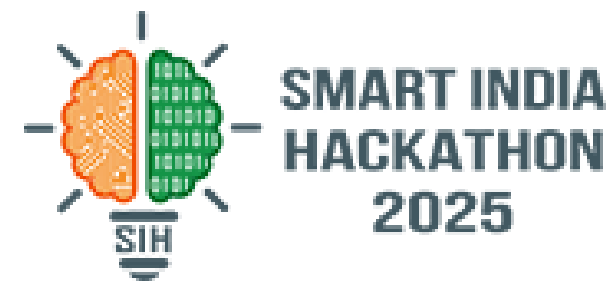
- WHO/BIS compliant Heavy Metal Pollution Index calculations
- Automated PDF generation with regulatory compliance

Environmental:

- Large-scale groundwater quality monitoring across regions
- Proactive pollution detection through predictive modeling



RESEARCH AND REFERENCES



References

Water Quality Standards:

- WHO Guidelines:

<https://www.who.int/publications/i/item/9789241549950>

- BIS 10500:2012:

<https://bis.gov.in/index.php/standards/technical-department/wrd/water-quality/>

Research Papers:

- Heavy Metal Pollution Index: Prasad & Bose (2001) - Environmental Geology
- ML for Water Quality: Gazzaz et al. (2012) - Marine Pollution Bulletin

Data Sources:

- Central Ground Water Board: <https://cgwb.gov.in/>
- Telangana Groundwater Data: <https://www.telangana.gov.in/departments/irrigation>

Technical Stack:

- Flask Framework: <https://flask.palletsprojects.com/>
- Scikit-learn Library: <https://scikit-learn.org/stable/>
- Leaflet Mapping: <https://leafletjs.com/>

Comparison with Existing Systems

Feature	GeoHMPI	Traditional Lab Testing	Government Monitring
Analysis Time	✓ 2 Seconds	✗ 3-7 Days	✗ 1-2 Weeks
Geographic Coverage	✓ Unlimited Locations	✗ Single Location	✗ Limited Points
Dynamic Real-Time	✓ Instant Predictions	✗ No	✗ Quarterly updates
Accessibility	✓ Any device/Internet	✗ Urban Labs Only	✗ Government Facilities
Technical Expertise	✓ Low(click-predict)	✗ High(Lab technicians)	✗ Government Facilities
Professional Reports	✓ Automated PDF	✗ Manual formatting	✓ Official Reports